

Process control

PRE-LAB PLAN

Unit Operations Lab 4

1/20/2026

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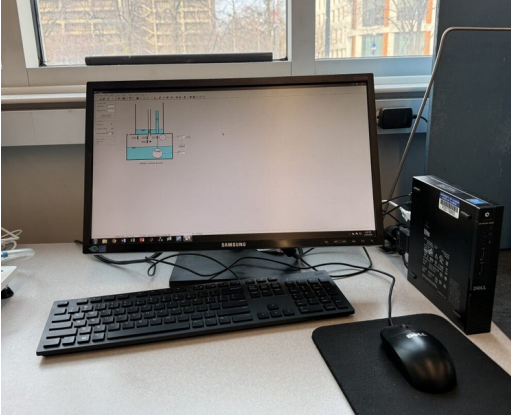
1. Experimental Objective

The purpose of the lab *Process Control: Level Control* is to experimentally determine how disturbances and changes to the controller configurations affect a dynamic process vessel. Experimental measurements are obtained by a single PID control loop and variables explored including pump speed, band values, outflow disturbances, setpoint changes and integral time.

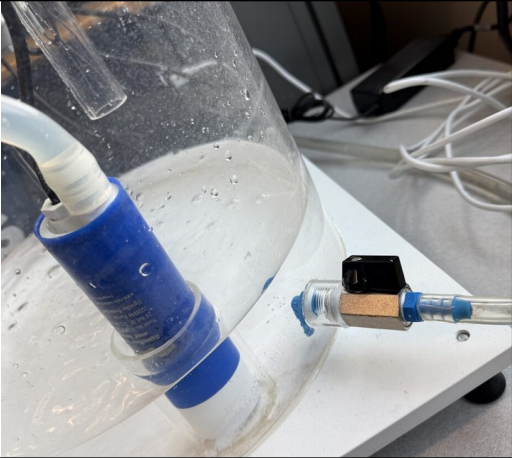
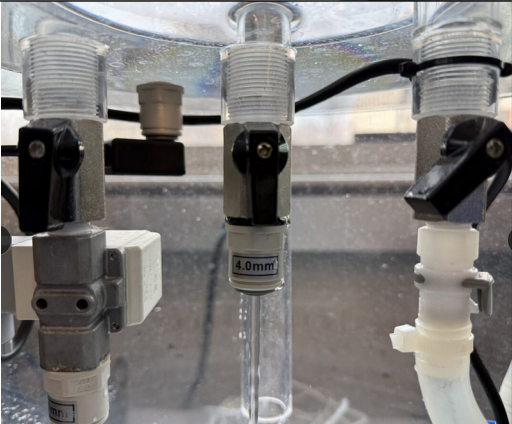
2. Experimental

This lab is divided into three experiments: Experiment 1 involves manual control of water level, Experiment 2 utilizes the PID control for a closed-loop system, in which only the Proportional Band (P) is used, and Experiment 3 also utilizes the PID control for a closed-loop system, in which both P and Integral Time (I) are used. Using the Armfield PCT50 Level Control, the main goal of each experiment is to collect relevant information and to plot water level versus time for different operating conditions. A summary of the apparatus is provided below.

2.1 Apparatus

Number	Name	Main vs Auxiliary	Description	Image
1	Computer	Main	Contains software to control pump and read outputs.	

				
2	Armfield PCT 50 level Control Experiment	Main	Vessel where process is carried out in the lower sump tank and the upper process	

3	Drain Valve	Auxiliary	Allows for the vessel to be drained	
4	Solenoid Valves	Main	Used to control flow of water to and from the upper process vessel	
5	Water Flow Meter	Main	Controls flowrate of water	

6	Pressure Sensor	Main	Reads pressure of water flowing out of upper process vessel, and helps to give water level reading in tank.	
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Table 1: Summary of the Apparatus

2.2 Material, Supplies & Additional Equipment

Materials/ Supplies	Material or supply	Description	Image
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DI Water	Material	De-ionized water	
Gloves	Supply	To cover skin and protect from potential contaminants.	
Timer	Supply	To time 5 minutes needed Experiment A.1.	

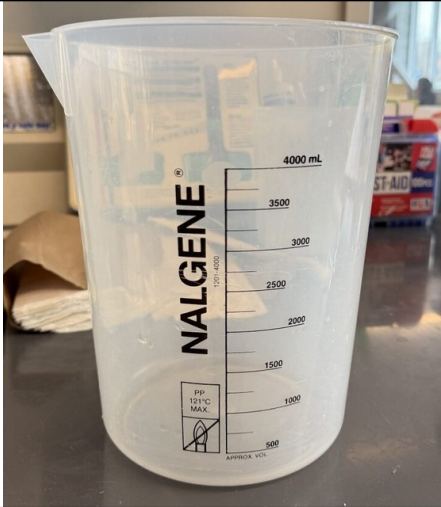
Beaker	Supply	To collect DI Water and fill vessel.	
Safety Goggles	Supply	Protects from contaminants entering/coming in contact with eyes.	

Table 2 : Summary of Materials and Supplies

2.3 Experimental Plan

Experiment 1: Manual Control of Water Level Open Loop:

Experimental Variables:

Independent Variables: Pump Speed, Solenoid valve state (open or closed)

Dependent Variables: Water Level (mm), fill/drain rate

Experimental Setup:

1. Turn on the computer, start the PCT50 software, and select experiment 1.
 - Ensure the virtual com port is enabled by the red-light indicator
2. Fill the sump tank with deionized water and fill to the correct level and fully open CV2
3. Initial adjustment for CV1. Check the readings on the mimic diagram and make sure that it reads zero. Proceed to fully open CV1 and set the pump speed to 50%. Also make sure that the mode of operation is set to manual

4. As the water level rises adjust CV1 until the level remains steady till it reaches mid height on the vessel

Experimental Procedure:

Initial Level Control

1. Starting with an empty process vessel. Slowly increase the pump speed.
2. Observe that the level stays constant if the pump matches the outflow. Record the pump speed at that level and press the “GO” icon to begin data collection
3. Record data for about 5 minutes then go to the next step while continuing the data collection

Disturbances

1. With the same pump speed and level values open the solenoid valve. Wait for the level to fall about 20mm. Note the pump speed and then increase pump speed to restore to the original level and let it run for a couple of minutes. Again, note the pump speed
2. Close the solenoid valve. The level should increase slowly as the flow now exceeds the outflow. When it has risen about 20mm note the pump speed. Then reduce the pump speed to restore the original level. After several minutes reduce the pump speed to 0%. Let the upper process vessel drain and let the data continue to collect. Save your data

Water fill and drain rate

1. Create a new results sheet. Then choose “GO” to start data collection
2. Gradually increase the pump speed to a value that is greater than in step 1.
3. Allow the level to increase to 100mm, from here stop the pump and let the vessel drain.
4. Once the pump is empty set the pump speed to 75% and let the vessel to fill again to 100mm. Then stop the pump and leave the vessel empty again. Stop data collection and export data into an excel file.

Data Analysis:

Graph the following results from the excel files:

Initial Level Control: Water level versus time at the 1st pump speed

Disturbances: Water Level Versus Time, solenoid on at 1st pump speed and 2nd pump speed, solenoid off at 2nd pump speed and 3rd pump speed

Water flow rate-drain and fill: Water level versus timing during filling (note the pump speed), Water level versus timing during draining (0% pump speed), Water level versus timing during filling (75% pump speed)

Describe what is happening in each experiment and why?

Report pump speeds needed to maintain the water level and the increase or decrease of the pump speed due to the different disturbances?

What were the water filling flow rates at the two different pump speeds?

Did the water draining flowrate stay the same or change?

Is it easy or hard to maintain constant water control with a manual valve?

Experiment 2: PID control of level, closed loop, P control only

Experimental Variables

Independent Variables: Proportional Band P, Setpoints (mm), Solenoid valve (open/closed)

Dependent Variables: Water Level (mm), Time, Stability, Offset which is the steady state error between actual level and setpoint.

Experimental Setup:

Follow the same experimental setup as experiment 1.

Procedure:

1. Choose the PID box on the mimic diagram and set the potential band (P) to 200%, the integral time I to 0, and the derivative time D to 0. Set the set point to 75mm. Click apply to enter the changes to the settings.
2. Choose automatic mode of operation and apply. The pump will start and the level will increase, and the water will drain from outlet valve CV2.
3. The level will continue to rise until it reaches a steady level. It may not correspond exactly to 75mm due to the offset when using a P-only.
4. Record the actual water level and note the stability of the water.
5. Open the solenoid valve to disturb the process. Observe the level falls, pump speeding up, with a large offset that remains

6. Once it has reached a somewhat steady state, note the stability of the level and the water level. Close the solenoid valve and allow the level to settle back to the original level height from step 1
7. Adjust the setpoint to 10mm and click apply. Observe the response to the increase in water level.
8. When it has settled to a constant value, note the stability, and return the setpoint to 75mm
9. The next 6 trials are going to repeat Steps 2-8. Using 6 different P values:
 - 100%, 50%, 20%, 10%, 5%, 2%
10. At the end of each trial ensure to select “STOP” and save the data to an excel file

Data Analysis:

Graph the following from results:

Water level versus time at the 7 proportional band values

Water level versus time for the increase outflow disturbances

Water level versus time for the change in setpoint (100mm)

Calculate the offset for each proportional band value, disturbance, and setpoint change and make a table to show the trend.

Discuss the stability of the water level at each P value

Choose a P value that gives the best stable control with a reasonably small offset. (will need this for the next experiment)

Experiment 3: PID control of level, closed loop, P + I control

Experimental Variables:

Independent Variables: Integral Time I, setpoint changes(75mm to 100mm), solenoid valves (open/close)

Dependent Variables: Water level, Time, resulting offset at each I value, Stability of the level

Experimental Setup:

Experiment should already be set up from experiment 2. If it has not, repeat the steps from experiment 1

Procedure:

1. Choose the PID box on the mimic diagram and set the proportional band (P) to the best value from experiment 2, integral time I to 0 sec, and the derivative time D to 0. Set the set point to 75mm and click apply to enter the changes
2. Choose automatic mode of operation. Select “GO” to begin collecting data. Let the water level stabilize to a constant level.
3. Once it has set the I value to 100 seconds
4. Observe the pump speed that varies to reduce the offset. When the level has settled to a constant value record the water level and the stability with the time.
5. Open the solenoid valve to disturb the process.
6. Observe the level falls slightly, the pump speed varies, and the level gradually returns to the set point.
7. Wait a couple of minutes to record the water level and the stability of the water level and the time.
8. Close the solenoid valve and allow the water level to return to the value from step 1.
9. Adjust the setpoint to 100mm.
10. Observe the increase of the water level. When it has settled record the water level and the stability then return the setpoint back to 75mm and observe the response
11. Repeat steps 2-10 with different I values starting at 50 seconds. Continue to reduce the integral time until the level becomes unstable
12. At the end of each trial select “STOP” and export the data onto an Excel File.

Data Analysis:

Graph the following from lab results

Water level versus time at different integral time and constant P values

Water level versus time for the increase outflow disturbance

Water level time for the change in setpoint (100mm)

Calculate the offset for each integral time value, disturbance, and setpoint change and make a table to show the trend.

Discuss the stability of the water level for each I value.

When does the water level control become unstable?

3. Project Safety Evaluation

Hazard	Yes/No	If Yes, Description
Potential electrical hazards?	Yes	Electrically operated pump and solenoid valve are in contact with water and pose an electrical hazard if a short circuit/failure occurs
Potential mechanical hazards?	Yes	Moving parts required for pump and solenoid-controlled valve both pose a mechanical hazard if they become damaged and are operated.
Potential pressure hazards?	Yes	The pump if operating outside of its standard conditions or the valve connecting the pump to the top process vessel
Potential temperature hazards?	No	
Hazardous/toxic chemicals involved	No	
Gloves required?	Yes	Gloves are always required for general lab safety purposes regardless of perceived hazard.
MSDS reviewed by all team members	Yes	Water (7732-18-5)
Process parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	25	
Pressure (psia)	25	
Safety controls	Yes/No	If yes, provided description
	Yes	

I have read relevant background material for the Unit Operations Laboratory entitled: “Process Control: Level Control” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

<u>Alexis Guillen</u>	<u>1/20/26</u>
Signature of team member	Date
<u>Khoa Nguyen</u>	<u>1/20/26</u>
Signature of team member	Date
<u>Rishabh S. Patel</u>	<u>1/20/26</u>
Signature of team member	Date
<u>Kyle Hankosky</u>	<u>1/20/26</u>
Signature of team member	Date

Miguel Piza

Signature of team member

1/20/26

Date